Atul Chaurasia

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Summary

AI/ML Engineer with 2+ years building production LLM systems, RAG pipelines and multi-agent applications. Currently building agentic AI for real-estate feasibility on **LangGraph**, **Temporal** and **MCP**, with evaluation suites reaching **precision and recall of 1.0** on real projects. Previously shipped an AI helpdesk handling **1,000+ requests a day** that cut resolution time by **60%**. Works across Python, FastAPI, PostgreSQL/pgvector and cloud deployment on GCP and AWS.

Education

Devi Ahilya Vishwavidyalaya (DAVV) — Indore, India
Master of Technology (M.Tech), Data Science — *2024 – 2026*

Shri Vaishnav Vidyapeeth Vishwavidyalaya — Indore, India
Bachelor of Technology (B.Tech), Computer Science — *2020 – 2024*

Work Experience

Data Scientist — February 2026 – Present
BestPeers Infotech — *Indore, India*

- Building an AI real-estate feasibility platform (**Python**, **FastAPI**, **LangGraph**, **Next.js**). It pulls live GIS and zoning data, computes the buildable envelope under **Miami 21** zoning and draws zoning-compliant building designs.
- Built the agent harness on **LangChain**, **LangGraph** and **Deep Agents** (skills + rules + tools + MCP) with **Temporal Cloud** durability: every LLM and tool call is an activity, and subagents run as child workflows. It powers **27 skills** across **5 agent-owned product screens**.
- Shipped the reference **Site & Context agent**: 8 tools, a LangGraph chat graph with **Gemini** tool calling and a grounding check, and zoning + GIS **MCP servers**. A verifier and **precision/recall evals** on recorded real projects reached **P = R = 1.0**, with end-to-end **LangSmith** tracing.
- Built the **Design agent**, a Deep Agents orchestrator with **8 specialist skills** (massing, floors, parking, core, facade, MEP). Each scheme is a Temporal workflow with preview, accept, undo. A check across **274 projects** cut zoning violations from **9 to 0** and raised the units drawn on 69 projects by **45%** (2,458 $\rightarrow$ 3,561).
- Built a zoning knowledge base on **PostgreSQL + pgvector** with **hybrid (vector + keyword) RAG**, plus a **Tesseract + PyMuPDF OCR pipeline** that indexed **461 pages of scanned zoning PDFs** that search couldn't read before.

AI/ML Engineer — September 2025 – February 2026
Me.AI — *Bangalore, India*

- Architected and deployed a production AI helpdesk processing **1,000+ daily requests** via a **32-node, 7-layer LangGraph StateGraph** with FastAPI, reducing L0 resolution time by **60%** and achieving **99.5% system uptime**.
- Engineered hybrid **RAG pipeline** combining Pinecone vector search, Tavily web retrieval, and LLM self-evaluation, achieving **85% query accuracy** across **8+ languages** for multilingual enterprise **natural language processing (NLP)** support.
- Developed and deployed **49 Model Context Protocol (MCP) tools** covering Azure AD administration, MQTT hardware diagnostics, and IT ticketing workflows, automating **200+ manual tasks per week** and enabling zero-shot issue resolution.
- Built **8-agent autonomous diagnostic system** with MQTT-based device operator monitoring **500+ endpoints**; **Human-in-the-Loop (HITL)** confirmation workflows reduced **Mean Time to Resolution (MTTR) by 40%**.
- Optimized LLM inference and vector retrieval data pipelines, improving p95 response latency and system throughput across thousands of concurrent daily production sessions.

Machine Learning Research Intern — August 2025 – November 2025
ICAR National Soybean Research Institute — *Indore, India*

- Engineered domain-specific features from **50,000+ labeled agronomic samples** and tuned an **XGBoost** ensemble model achieving **86% disease prediction accuracy**, reducing false negatives by **12%**.

Machine Learning Intern — March 2024 – July 2024
Supersourcing — *Indore, India*

- Built production **NLP** resume parser using **spaCy Named Entity Recognition (NER)**, Regex, and **BERT** embeddings achieving **92% extraction accuracy**; deployed as a **FastAPI** REST API microservice.
- Developed **LLM**-based text summarization pipeline using **LLaMA-2 7B**, reducing recruiter manual screening time by **70%** and increasing candidate throughput by 5$\times$.
- Processed **600,000+ job listings**; trained **XGBoost** salary prediction model achieving **R^2 = 0.86** for data-driven compensation benchmarking.

Projects

Traffic-GNN: Spatio-Temporal Traffic Prediction | *Python, PyTorch, GAT, LSTM* GitHub

- Built **Graph Attention Network (GAT) + LSTM** deep learning model on a **66,000-node** OpenStreetMap graph for multi-horizon traffic prediction; implemented A* routing with dynamic edge weights derived from predicted congestion.

Deep Learning Architectures from Scratch | *Python, PyTorch, NumPy* GitHub

- Implemented **10+ core architectures** from scratch - Transformer, BERT, LLaMA 2, ViT, LoRA, QLoRA, GAN, VAE, matching reference implementations; published as open-source **machine learning** resource.

Airline Multi-Agent AI Assistant | *Python, Google ADK, Multi-Agent Systems* GitHub

- Built **4-agent orchestration system** for end-to-end airline booking workflows with stateful conversation management; validated across **500+ automated test scenarios** covering edge cases and re-booking flows.

YouTube Semantic Search Engine | *Python, RAG, Pinecone, Sentence Transformers* GitHub

- Developed semantic search over **10,000+ YouTube transcripts** achieving **78% retrieval precision** at sub-**200ms** latency using dense vector embeddings and approximate nearest-neighbor **(ANN)** search.

Technical Skills

Programming Languages: Python, SQL

LLM & Agent Frameworks: LangChain, LangGraph, Google ADK, Model Context Protocol (MCP), LangSmith, Temporal

AI & Machine Learning: LLMs, RAG, NLP, Deep Learning, Prompt Engineering, Fine-Tuning, Agent Orchestration, HITL Workflows, Ensemble Learning, Model Evaluation, Hyperparameter Tuning

ML Libraries: PyTorch, Transformers, BERT, ViT, XGBoost, spaCy, Scikit-learn, CNN, LSTM, GAN, VAE, LoRA, QLoRA

Backend & APIs: FastAPI, REST API Development, Microservices Architecture, System Design

Databases & Vector Stores: PostgreSQL, pgvector, MongoDB, Pinecone, Neo4j

Cloud & DevOps: AWS, GCP, Docker, Git, CI/CD, Model Deployment, MLOps, Data Pipelines

Certifications

DeepLearning.AI — Neural Networks and Deep Learning (Coursera)

Accenture — Data Analytics and Visualization

Achievements

- Solved **120+ algorithmic problems** on LeetCode covering arrays, graphs, dynamic programming, and system design.
- Grew **Twitter/X to 6,000+ followers** by sharing AI/ML content on LLMs, RAG, and agentic AI.